Belize Climate Risk Assessment: Inception Report

IRI/NCDP, Columbia University

2025-12-26

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Executive Summary

This Inception Report outlines the detailed methodology, tools, and workplan for the Belize climate risk assessment project. The report consolidates the approach for data collection, analysis, and community engagement, building on the desk review findings presented in the Situational Analysis Report. This document provides:

- Detailed methodology including tools for data/information gathering
- Quality assurance mechanisms
- Data/information collection processes
- Analysis methodologies

- Fieldwork plan (to be finalized after discussions with NMS and NEMO)
- Final Workplan with Gantt Chart

Detailed Methodology

Tools for Data and Information Gathering

The project will employ a combination of technical tools and participatory methodologies to gather comprehensive data and information:

Technical Data Collection Tools:

- **Earth Observation and Satellite Data:** Google Earth Engine (GEE) for accessing and processing satellite datasets, including precipitation, temperature, soil moisture, and other climate variables
- **Climate Data Tools (CDT):** IRI's software package for quality control of climate data
- **Climate Predictability Tool (CPT):** User-friendly platform for producing tailored seasonal climate forecasts
- **ENACTS Initiative:** Methods, tools and training for developing high-resolution climate datasets and Maprooms
- **Geographic Information Systems (GIS):** For spatial analysis and mapping of hazards, exposure, and vulnerability

Participatory Data Collection Tools:

- **Farmer Focus Group Methodologies:** Structured community discussions to elicit historical risk years
- **Mobile Crowdsourcing Tools:** WhatsApp-based tools for scalable data collection
- **Interactive Educational Exercises:** Games and simulations to understand household-level decisions
- **Data Cleaning and Reconciliation Tools:** Centralized platforms for quality-checking crowdsourced data

Quality Assurance Mechanisms

Data Quality Control:

1. **Automated Quality Checks:** Use IRI Climate Data Tools (CDT) for systematic quality control of climate data, including:
   - Detection of outliers and missing values
   - Consistency checks across datasets
   - Temporal and spatial validation
2. **Multi-Source Validation:** Cross-validate data from multiple sources:
   - Compare satellite data with ground-based observations
   - Validate community recall against historical climate records
   - Cross-reference ministry consultations with technical assessments
3. **Community Data Quality Assurance:**
   - Visualize crowdsourced data on historical risk years over multiple spatial scales
   - Identify and correct probable errors through centralized quality-checking tools
   - Integrate multiple observational datasets and qualitative data using information theory-based methodologies
4. **Expert Review:** Regular review by project team and stakeholders to ensure data quality and relevance

Data and Information Collection Processes

Desk Review Process:

- Review existing data, information, maps and studies for the project area
- Conduct gap assessment of previous risk and vulnerability assessments

- Compile comprehensive data inventory (see Data Inventory section below)
- Analyze existing forecast products and data availability

Stakeholder Consultation Process:

- Conduct consultations with government ministries and agencies (see Ministry Consultations section below)
- Engage with National Meteorological Service (NMS) and National Emergency Management Organisation (NEMO)
- Identify priority hazards, preferred datasets, and community validation capacity
- Document stakeholder requirements for forecasts and warnings

Community Engagement Process:

- Identify priority use cases for community validation
- Conduct Farmer Focus Groups using IRI's standardized methodology
- Employ multiple engagement modalities (in-person focus groups, mobile crowdsourcing, interactive exercises)
- Validate metrics and decision rules through community engagement
- Fill data gaps where official monitoring networks are sparse

Analysis Methodologies

Hazard Analysis:

- Analyze hazard characteristics (intensity, frequency, probability) using historical data
- Evaluate climate variables and climate change scenarios
- Develop hazard maps linking forecasts and vulnerabilities
- Use GEE maprooms to visualize hazard, exposure and vulnerability layers

Vulnerability and Exposure Analysis:

- Conduct community vulnerability analysis for relevant natural hazards
- Analyze community practices for natural resources management
- Estimate potential human and economic losses based on exposure and vulnerability
- Assess factors affecting community capacity for risk and impact assessments

Forecast Integration:

- Co-identify relevant weather and climate parameters
- Establish critical threshold values for hazard warnings
- Develop understandable, recognizable, and timely warnings for NMS and NEMO
- Link forecasts with vulnerabilities for the project area

Ministry Consultations

This section summarizes consultations conducted with government ministries and agencies regarding priority natural hazards, preferred datasets, and capacity for community validation.

Ministry Contacts

Ministry	Contact
Hydrology (NHS)	Tennielle Hendy; chief@nhs.gov.bz
Tourism	Darcy Correa; darcy.correa@tourism.gov.bz

Ministry	Contact
Coastal Zone Management (CZMAI)	Mrs. Arlene Young (director@coastalzonebelize.org); Mr. Isreal Correa (gismanager@coastalzonebelize.org); Ms. Chelsea Perrera (coastalplanner@coastalzonebelize.org)
Livestock	William Usher; ceo@belizelivestock.org
Forestry	Wilfredo Zetina; ZetinaW@gobmail.gov.bz
Sustainability / Climate Change	Judene Tingling-Linares; coord.pppu@environment.gov.bz

Ministry Priorities and Capacity

Ministry	Priority Hazards	Preferred Datasets	Community Validation Capacity
Hydrology (NHS)	- Floods (all types); - Drought	- Twice daily river levels for floods; - Coastal environment data for coastal flooding (needed); - Groundwater levels data nationally for hydrological drought (needed)	Limited due to human resources; can execute where suitable. Primary contact: Tennielle Hendy. Community members read staff gauges; annual training sessions conducted.
Tourism	- Hurricane; - Flood; - Drought; - Heat waves; - Lightning Storms	- Relevant weather station data country wide; - Gauging stations along rivers	Not specified
Coastal Zone Management (CZMAI)	**Long-term:** - Increasing sea surface temperature; - Sea level rise. **Short-term:** - Hurricanes/storm surges	- Global datasets on sea temperature rise from satellite sensors; - Sea level data from sea loggers/satellite altimetry/tide gauges; - Hurricane and storm surge forecasts with at least 3 days' notice	CZMAI has capacity and is willing to aid in community validation for risk analysis
Livestock	1. Flood; 2. Heat Stress Index (Temperature & Humidity)	- Rain Intensity; - Temperature and Humidity	No capacity available
Forestry	- Wildland fires; - Drought; - Hurricanes	- Any relevant datasets that are available	Contact person: Wilfredo Zetina. Would need assistance from team at main office (only two officers at San Ignacio station)

Ministry	Priority Hazards	Preferred Datasets	Community Validation Capacity
Sustainability / Climate Change	1. Flooding [Rainfall; discharge, soil moisture]; 2. Tropical cyclones; 3. Wildfires [thermal hotspots/satellite imagery]; 4. Drought (Standardized Precipitation Indices, SPIs)	- Rainfall and discharge data for flooding; - Soil moisture data; - Thermal hotspots/satellite imagery for wildfires; - Standardized Precipitation Indices (SPIs) for drought monitoring	Capacity exists in different departments; human resource limitation to actively participate might be challenging

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NMA Forecast Products

Information in this section is based on desk review of NMA forecast products and consultation notes.

The National Meteorological Agency (NMA) provides forecast products, though their website reliability is a concern. The NMA is the primary government agency responsible for meteorological services in Belize:

1. **Seasonal Forecast:** IRI-style seasonal forecast available
2. **Agricultural Forecast:** Shorter-term agricultural forecast available
3. **Common Alerting Protocol (CAP):** Belize is currently implementing this common standard for short-term disaster alerts via SMS. CAP is an international standard format for exchanging emergency alerts and public warnings.

Limitations and Data Access Challenges:

1. **Website reliability issues:** The NMA website has been noted as unreliable, which poses challenges for accessing current forecast information.

2. **Historical forecast data access:** There is no straightforward way to access historical forecast information from the NMA. This limitation affects the ability to:

   - Validate forecast skill and accuracy over time
   - Conduct retrospective analyses of forecast performance
   - Develop forecast-based action triggers based on historical patterns

3. **Alternative data sources:** Given these limitations, the following alternative approaches should be considered:

   - **Public forecast products:** Explore publicly available forecast products from international sources that may provide historical data and verification records
   - **Collaborative research:** Leverage ongoing collaborative research on NextGen forecasting methodology that may provide alternative forecast data sources or historical records
   - **Direct engagement:** Continue engagement with NMA to explore potential data sharing agreements or alternative access methods for historical forecast information

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Data Inventory

This section presents the comprehensive data inventory compiled during the desk review process, identifying candidate datasets for the climate risk assessment and maptool development.

The following datasets have been identified for the Belize climate risk assessment project. These datasets are organized by category: **Hazard**, **Exposure/Vulnerability/Impact**, and **Forecasts/Projections**. This inventory serves as a foundation for:

- Identifying candidate datasets for the design dashbboard development
- Understanding data availability and gaps across hazard, exposure, and vulnerability dimensions
- Informing data acquisition and integration strategies
- Supporting community validation efforts by identifying where local information can fill data gaps

The data inventory is generated from an Excel workbook containing multiple sheets. Each category is processed and displayed as a formatted table below. The inventory includes datasets relevant for:

- **Hazard assessment:** Datasets for measuring and monitoring priority natural hazards identified through ministry consultations
- **Exposure and vulnerability:** Datasets for understanding what and who is at risk, including population, infrastructure, and socioeconomic indicators
- **Forecasts and projections:** Datasets for climate risk assessment and future climate scenarios

Hazard Datasets

Name	Type	Historical Availability	Temporal Resolution	Spatial Resolution	Data Latency	Hazard Metrics	Comments
CHIRPS	Rainfall	1981+	5-day (main), 10-day, 1-day	5.5 km	3 weeks	Season onset & demise	Best suited for measurements of drought and rainy season timing
MODIS NDVI	Vegetation	2000-2023	8-day, 1-day, 16-day	250m	<24hrs	Vegetation greenness index	Best suited to measuring agricultural drought and land degradation
VIIRS NDVI	Vegetation	2012+	8-day, 1-day, 16-day	500m	<24hrs	Vegetation greenness index	Post-2023 supplement to MODIS.

Name	Type	Historical Availability	Temporal Resolution	Spatial Resolution	Data Latency	Hazard Metrics	Comments
MODIS NDWI	Flooding	2000-2023	8-day, 1-day, 16-day	250m	<24hrs	% flooded area	Best suited to measuring flooding from rivers and storm surge. Note that the Global Flood Database provides an inventory of validated major flood events as measured by MODIS.
VIIRS NDWI	Flooding	2012+	8-day, 1-day, 16-day	500m	<24hrs	% flooded area	Post-2023 supplement to MODIS.
ERA5 / GLDAS	Temperature	1979+	1-day	27km	<24hrs	# days above temperature danger threshold	Reanalysis data (i.e., based on global climate models).
CHIRTS	Temperature	1981+	5-day (main), 10-day, 1-day	5.5 km	6 days	# days above temperature danger threshold	Combination of station data and ERA5
PML V2 Evapotranspiration	Drought	2000+	8-day	500m	Annual	Gross primary production	Modeled evapotranspiration and soil moisture

Name	Type	Historical Availability	Temporal Resolution	Spatial Resolution	Data Latency	Hazard Metrics	Comments
IBTRACS	Tropical storms	1980+	1-hour (typical)	1km		Tropical storm tracks	Used to provide information on incidence, duration and itensity of tropical storms.

Exposure, Vulnerability, Impact Datasets

Name	Type	Application	Comments
Global Human Modification	Index of environmental stressors	Exposure	Indexes built-up area, extractive industries, infrastructure
VIIRS Night Lights	Index of economic activity	Exposure	Estimates spatial density of economic activity using remotely sensed night lights
WorldPop Residential Population Density	Gridded population data	Exposure	Uses machine learning to provide high-res population estimates
Copernicus Land Cover	Land use	Exposure	Remotely sensed probabilistic estimate of land cover material
Koppen Climate Classification	Agroecological zones	Exposure	Based on average climate conditions; dataset also includes soil type
NASA SRTM Digital Elevation	Terrain elevation	Exposure	Used in conjunction with coastline dataset to estimate sea level rise scenarios
HydroSHEDS	River extent and flow; flood plains	Exposure	Used to provide information on (1) Areas that currently lie in flood plains and (2) Areas that are likely to lie in a flood plain in the future after accounting for sea level rise.
Landsat Global Forest Cover Change	Estimated forest canopy cover	Vulnerability	Based on Landsat; updates every 5 years
World Database on Protected Areas	Inventory of designated conservation areas	Vulnerability	
Oxford MPI	Multidimensional poverty index	Vulnerability	Available by administrative area

Name	Type	Application	Comments
CARICOM / WFP Real-time Food Insecurity Monitoring	Food insecurity	Impact	Updates daily during 3-month collection intervals. Includes WFP food insecurity indicators - FCS, rCSI, HHS
EMDAT	Historical disaster events	Impact	Georeferenced news-based event data 1960-2019 (GDIS dataset contains detailed geocodes where available)
Dartmouth Flood Observatory	Historical disaster events	Impact	Georeferenced news-based event data 1985-present
NASA Goddard Global Landslide Catalog	Historical disaster events	Impact	Georeferenced historical landslide impacts 1970-2019
Global Fire Atlas	Historical disaster events	Impact	Georeferenced news-based event data 2003-2016
World Bank Climate Change Knowledge Portal	Historical disaster events	Impact	Georeferenced news-based event data 1980-present
Community data on livelihood calendars and climate impacts			To be collected in 2026 to supplement publically available data sources, and help design locally relevant impact-based forecast triggers

Forecasts, Projections Datasets

Name	Type	Historical Availability	Temporal Resolution	Spatial Resolution	Data Latency	Hazard Metrics	Comments
Belize NMA Forecast Products							To be populated by NMA

Name	Type	Historical Availability	Temporal Resolution	Spatial Resolution	Data Latency	Hazard Metrics	Comments
IRI Global Seasonal Forecasts	Temperature and precipitation	1997+	3-month	100km	Once per month	Precipitation and temperature totals	Available in flexible format - i.e., can be tailored to different levels of predicted event severity. Downscaled and locally tailored versions of this type of forecast are being produced as part of Diego's effort.
IRI Subseasonal Forecasts	Temperature and precipitation	1997+	4-week	100km	Weekly	Precipitation and temperature totals	Experimental supplement to seasonal forecast
CHC Subseasonal Forecasts	Temperature and precipitation	1999+	15-day, 30-day	100km	Weekly	Precipitation and temperature totals	
ECMWF Medium-Term forecast	Temperature, precipitation, wind	1957+	10-15 days	10km	Daily	Precipitation and temperature totals	Only some spatial data is publically available, and is not readily available in an interoperable format. However, charts and maps of forecasts are published regularly.

Name	Type	Historical Availability	Temporal Resolution	Spatial Resolution	Data Latency	Hazard Metrics	Comments
NMME Seasonal Forecast	Temperature and precipitation	1997+	3-month	100km	Once per month	Precipitation and temperature totals	
NOAA NHC Hurricane Forecast	Hurricanes	2008+	5 days	Varies	Varies	Hurricane tracks, wind fields, storm surge estimates	

NASA NEX-DCP30	Rainfall and Temperature	1900-2006; 2006-2099, (projections)	Monthly	27km		% future change in monthly rainfall, temperature	Allows for consideration of different emissions scenarios. For each scenario, the median and 25th-75th percentiles across various climate models are presented. Note that this does NOT account for uncertainty due to other factors that influence the natural variability of climate, like the El Nino cycle. For that, the climatology analysis can provide context.

Community Information Integration

This section outlines the approach for incorporating community information into the risk analysis, based on desk review findings, ministry consultation feedback, and IRI methodologies for climate risk assessment.

IRI Methodologies for Community Engagement

The integration of community information will leverage IRI's standardized methodologies that have been successfully applied at scale in multiple countries:

Farmer Focus Group Methodologies: IRI's participatory approach employs structured community discussions to elicit historical risk years from farmers. This methodology enables communities to rank their worst disaster years for specific hazards (e.g., drought, flooding), which can then be compared against climate data to validate forecast decision rules. For example, a drought action rule meant to trigger during a 1-in-5-year drought can be validated against the worst years farmers recall from the past 40 years. The methodology is based on consensus-building among community members, ensuring that the final result represents collective local knowledge.

Interactive Educational Exercises: IRI's approach includes "games" or simulations that demonstrate potential household-level decisions that climate risk assessment can inform, while simultaneously understanding farmers' typical production and investment constraints. These exercises help elucidate the theory of change for climate risk assessment and check its plausibility: for example, understanding whether farmers would make productive investments (like improved seeds) if climate risk information could inform input decisions before a drought year.

Mobile Crowdsourcing Tools: In cases where financial, logistical, or safety constraints limit the feasible scale of in-person focus groups, IRI has developed mobile tools leveraging commonly used messaging platforms like WhatsApp. These tools use behavioral design elements and incentives to elicit reliable responses from farmers, enabling scalable data collection. These tools have been successfully piloted with hundreds of farmers and include dashboards that allow non-technical users to deploy, customize, and monitor survey campaigns.

Data Cleaning and Reconciliation Tools: Given that community-generated data is subject to entry errors, inaccurate recall, and other complexities, IRI has developed tools to help local users visualize crowdsourced data on historical risk years over multiple spatial scales, identify probable errors, and enter corrections via a centralized ledger. Additionally, IRI has developed a methodology based on information theory for integrating multiple observational datasets and qualitative data (like farmers' recall) into a holistic assessment of historical drought tendency.

Seasonal Assessment Methodology: A critical component of climate risk assessment is continual monitoring throughout the season to assess policy performance, identify gaps, and reconcile stakeholders' expectations. IRI's seasonal assessment methodology draws on multiple sources of data, including biophysical measurements, crowdsourced reports from farmers, and humanitarian food security monitoring, presenting stakeholders with structured questions to gather informed assessments of seasonal conditions.

IRI Tool Platforms and Resources:

- **Forecast Action Planning Tools:** Interactive map tools for visualizing forecasted disaster severity and evaluating decision rules (example: http://iridl.ldeo.columbia.edu/fbfmaproom2/ethiopia-ond). These tools allow users to adjust outcome types and severity thresholds, incorporate observational data, and evaluate historical performance.
- **Data Cleaning Tools:** Centralized platforms for quality-checking crowdsourced data (example: https://fist-cleandat.iri.columbia.edu/comzambia). These tools enable visualization of historical risk years over multiple spatial scales and error correction via centralized ledgers.
- **Decision Support Tools:** Interactive tools for developing and evaluating AA decision rules, including examples from index insurance applications (https://fist-shiny.iri.columbia.edu/shiny_apps/Apps/Zambia_Window_c
https://fist-shiny.iri.columbia.edu/shiny_apps/Apps/zambia_optimization_app/).
- **AA Training Materials:** Comprehensive training resources for AA trigger development (example: https://fist.iri.columbia.edu/publications/docs/Lesotho_FbF_Material2022/ - username: FBFLesotho, password: resilience).
- **Next Generation Drought Index (NGDI):** Participatory tool for co-generation and evaluation of satellite datasets and insurance solutions (https://bit.ly/WBIRISenOpt - username: WB, password: IRI).
- **Seasonal Assessment Resources:** Methodology documentation and examples (https://fist.iri.columbia.edu/publicati

- **IRI Climate Data Tools (CDT):** Software package for quality control of climate data (https://iri.columbia.edu/our-expertise/climate/tools/cdt/).
- **IRI Climate Predictability Tool (CPT):** User-friendly platform for producing tailored seasonal climate forecasts (https://iri.columbia.edu/our-expertise/climate/tools/cpt/).
- **ENACTS Initiative:** Methods, tools and training for developing high-resolution climate datasets and Maprooms (https://iri.columbia.edu/resources/enacts/).

Integration Strategy

The community information integration process will follow an enhanced approach that incorporates IRI methodologies:

1. **Identify relevant use cases:** Identify priority use cases that would benefit from community validation and local knowledge integration, focusing on hazards where community input can fill critical data gaps or validate technical assessments.
2. **Historical risk event validation:** Conduct Farmer Focus Groups using IRI's standardized methodology to elicit community-identified historical disaster years. This information is critical for retrospective evaluation of climate risk assessment decision rules, ensuring that forecast triggers align with community experiences of actual disaster events.
3. **Gather community information:** Employ multiple engagement modalities:
   - **In-person focus groups:** Structured community discussions in accessible locations with special consideration for indigenous communities in Toledo requiring culturally-appropriate engagement
   - **Mobile crowdsourcing:** WhatsApp-based tools for areas where in-person focus groups are constrained by logistics, safety, or financial limitations
   - **Interactive educational exercises:** Games and simulations to understand household-level decisions and validate the theory of change for climate risk assessment interventions
4. **Validate metrics and decision rules:** Validate risk metrics, hazard indicators, and forecast decision rules through community engagement:
   - Compare technical assessments against local experiences and observations
   - Validate that climate risk assessment triggers would have captured known historical disaster events
   - Ensure decision rules align with community-identified worst years for each hazard
   - Use community input to refine trigger thresholds and action protocols
5. **Fill data gaps:** Frame community information collection as a means to fill gaps in existing data sources, particularly where:
   - Official monitoring networks are sparse (e.g., groundwater levels for hydrological drought, upper catchment river gauges)
   - Historical records are incomplete (e.g., EMDAT data limitations for Belize)
   - Local impacts and vulnerabilities are not captured in national datasets
   - Community-specific adaptation strategies and capacities need documentation
6. **Data quality assurance:** Implement IRI's data cleaning and reconciliation processes:
   - Visualize crowdsourced data on historical risk years over multiple spatial scales
   - Identify and correct probable errors through centralized quality-checking tools
   - Integrate multiple observational datasets and qualitative data using information theory-based methodologies
7. **Seasonal monitoring:** Establish continual monitoring protocols using IRI's seasonal assessment methodology, drawing on biophysical measurements, crowdsourced farmer reports, and humanitarian food security monitoring to assess climate risk assessment policy performance throughout the season.

Technology Platforms and Decision Support

Decision Support Tools Integration: Community input will directly inform the development and evaluation of climate risk assessment decision rules using IRI's decision support methodology. This includes: - Creating an inventory of threshold/trigger methodologies used by stakeholders to enhance harmonization - Generating lists of known historical disaster events for each region and sector to validate decision rule performance - Developing trainings for applicable agencies on the use of triggers and impact-based thresholds - Creating learning notes that explain the complete trigger/threshold development process

Participatory Validation: Community recall of historical disaster years will be systematically compared against climate data to validate forecast decision rules. This participatory validation ensures that technical climate risk assessment triggers capture events that communities recognize as disasters, reducing basis risk and increasing community trust in the system.

Multi-Source Integration: Community data will be integrated with: - Biophysical measurements (satellite data, gauge readings, weather station data) - Humanitarian food security monitoring (IPC/VAM assessments) - Government administrative data - Expert knowledge from ministry consultations

This multi-source approach enables a holistic assessment of historical risk tendency and strengthens the foundation for climate risk assessment decision rules.

Fieldwork Plan

This section will be finalized after discussions with the National Meteorological Service (NMS) and National Emergency Management Organisation (NEMO).

The fieldwork plan will include:

- **Community engagement locations:** Identification of priority communities for focus groups and participatory activities
- **Stakeholder consultation schedule:** Coordination with NMS, NEMO, and other key stakeholders
- **Data collection timeline:** Schedule for community validation activities, focus groups, and mobile crowdsourcing campaigns
- **Logistical considerations:** Access, safety, and cultural considerations for fieldwork activities
- **Community validation protocols:** Detailed procedures for conducting focus groups, games, and mobile data collection

Note: The detailed fieldwork plan will be developed in coordination with NMS and NEMO following the inception phase.

Final Workplan with Gantt Chart

This section presents the final workplan with a Gantt chart showing project tasks, timeline, and delivery schedule based on the project deliverables.

Task Summary Table

Task	Due Date	Status
Review exising data, information, maps and studies for the project area and conduct a gap assessment of previous risk and vulnerability assessments.	2025-11-14	Completed

Task	Due Date	Status
Evaluate climate variable and climate change (CC) scenarios: Analyse hazard characteristics(e.g. intensity, frequency and probability) of	2025-12-26	Completed
identified hazards taking historical data into consideration;	2025-12-26	Completed
Conduct an assessment to collect baseline information and identify gaps relating to the impacts of different weather hazards and the exposure of vulnerable assets and population.	2025-12-26	Completed
Conduct community vulnerability analysis for relevant natural hazards, considering historical data sources and potential future hazard events in the vulnerability analysis.	2026-01-30	Deliverables
Analyse community practices for natural resources management:	2026-01-30	Deliverables
Analyse factors affecting the community capacity to undertake risk and impact assessments, to plan and implement mitigation and adaptation actions.	2026-01-30	Deliverables
Estimate potential human and economic losses based on the exposure and vulnerability of people (particularly youth, elderly, persons with disabilities) buildings and infrastructure.	2026-02-27	Deliverables
Co-Identify relevant weather and climate parameters and establish critical threshold values required to issue a warning for hazard events	2026-02-27	Deliverables
Forecast-related deliverables	2026-03-31	Deliverables
Develop hazard map tools linking forecasts and vulnerabilities for the project area.	2026-03-31	Deliverables

Gantt Chart

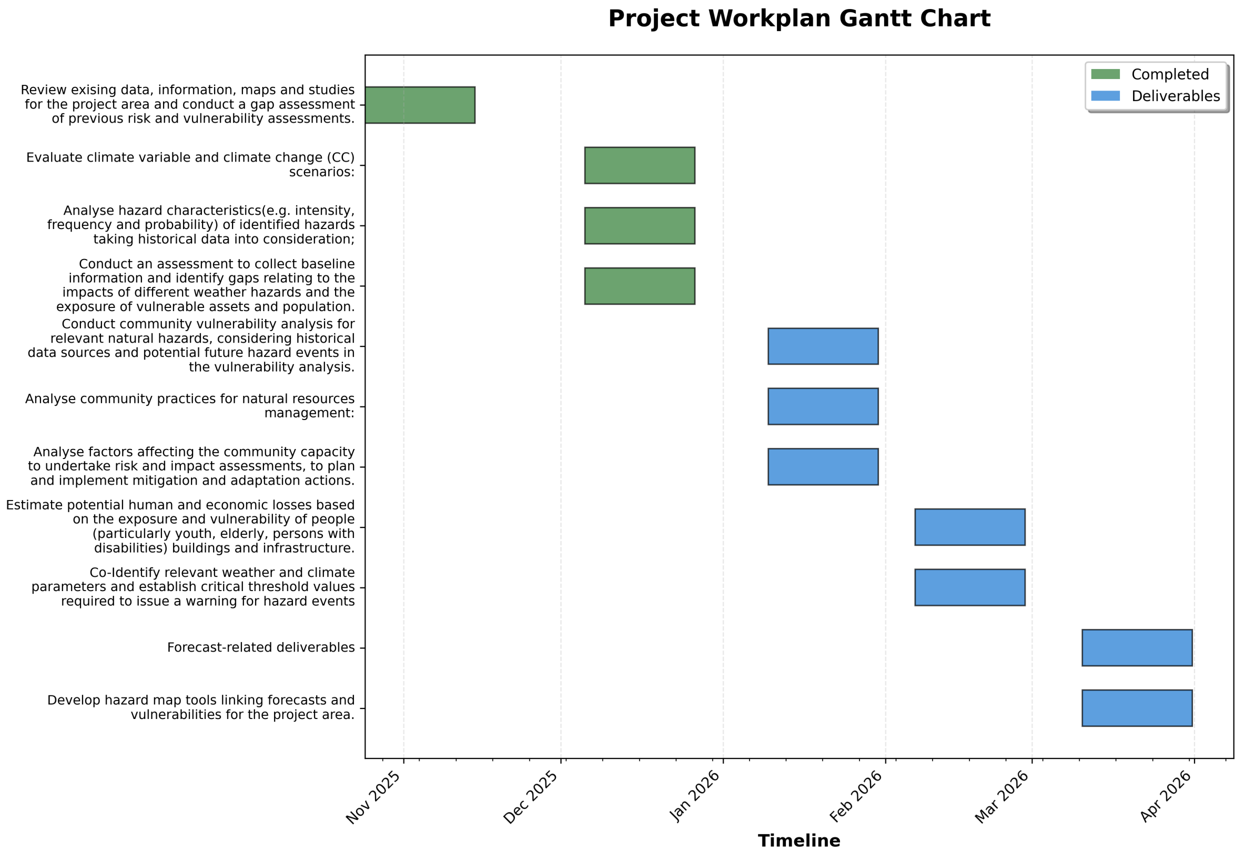

Figure 1: Project Workplan Gantt Chart

References

Consultation Sources

Ministry Consultations (2024):

- **Hydrology (NHS):** Consultation with Tennielle Hendy (chief@nhs.gov.bz)
- **Tourism:** Consultation with Darcy Correa (darcy.correa@tourism.gov.bz)
- **Coastal Zone Management (CZMAI):** Consultations with Mrs. Arlene Young (director@coastalzonebelize.org), Mr. Isreal Correa (gismanager@coastalzonebelize.org), and Ms. Chelsea Perrera (coastalplanner@coastalzonebelize.org)
- **Livestock:** Consultation with William Usher (ceo@belizelivestock.org)
- **Forestry:** Consultation with Wilfredo Zetina (ZetinaW@gobmail.gov.bz)
- **Sustainability / Climate Change:** Consultation with Judene Tingling-Linares (coord.pppu@environment.gov.bz)

Data Sources

- **Belize Data Inventory:** Excel workbook containing comprehensive dataset inventory (Belize Data Inventory.xlsx)
- **EMDAT (Emergency Events Database):** Historical disaster records (noted as sparse for Belize)
- **World Bank:** Climate risk data and assessments

- **Climate Risk Profile:** Assessment of communities' adaptive capacity and adaptation strategies

Climate Data Sources

- **World Bank Climate Change Knowledge Portal (CCKP):** Climate context information, visualizations, and analyses for Belize. Available at: https://climateknowledgeportal.worldbank.org/country/belize
- **NOAA ETOPO Global Relief Model:** Topographic elevation data. Available at: https://www.ncei.noaa.gov/produ global-relief-model
- **Gridded Population of the World, Version 4 (GPWv4):** Population density data. Available at: https://cmr.earthdata.nasa.gov/search/concepts/C1597159135-SEDAC.html

Forecast Products

- **National Meteorological Agency (NMA), Belize:** Seasonal forecasts, agricultural forecasts, and Common Alerting Protocol (CAP) implementation

Desk Review Notes

- Consultation notes and findings compiled during the desk review process (Belize desk review notes.txt)

Climate Risk Analysis Plan Sources

Citations from Section 1 (Executive Summary):

1. Climate Risk Profile: Belize - International Institute for Sustainable Development, accessed December 15, 2025, https://www.iisd.org/system/files/2025-04/belize-climate-risk-profile.pdf

2. Belize Bolsters Climate and Disaster Resilience with the World Bank's Support, accessed December 15, 2025, https://www.worldbank.org/en/news/press-release/2025/07/29/belize-bolsters-climate-and-disaster-resilience-with-the-world-bank-s-support